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Measuring Chronic Rheumatic Pain in Mexican Americans: Cross-Cultural Adaptation of the McGill Pain Questionnaire

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ABSTRACT. We performed a cross-cultural adaptation of the McGill Pain Questionnaire (MPQ) from English to Spanish for studying Mexican Americans in South Texas. Each of the 78 single-word pain descriptors in the original MPQ was translated into Spanish by a panel of nine bilingual health researchers, preserving the original structure of the questionnaire. The pain-intensity content (PIC) of the words in each language was then rated on a 100 mm visual analog scale by 8 bilingual health care providers and 10 bilingual health-care consumers. The correlation between Spanish and English average PIC ratings was strong ($r = 0.85$ for providers, $r = 0.80$ for consumers). The translated Spanish version was compared to the original English in a group of 50 bilingual Mexican-American patients with musculoskeletal pain, who completed the MPQ in both languages. There was no difference in Average Pain Rating Index between the Spanish and English versions (29.8 ± 14.7 vs 29.1 ± 15.8, $p = 0.55$), and agreement between the two language versions was almost perfect ($r_i = 0.85$). Test–retest reliability was measured in two groups of hospitalized patients (25 per group), one composed of monolingual Spanish speakers and the other of monolingual English speakers. Each subject completed the MPQ, the McGill Pain Map, two 10-cm visual analog scales measuring pain now and within the past week, the bodily pain items of the MOS-SF36 survey, and the Modified Health Assessment Questionnaire, on two occasions one day apart. Test–retest reliability of the Spanish and English components of the MPQ was not significantly different and was comparable to that of the other pain and health status instruments. We conclude that the Spanish MPQ is cross-culturally equivalent to the original English and has similar concurrent validity and reliability. This questionnaire is suitable for cross-cultural studies of pain comparing Spanish-speaking Mexican Americans with English-speaking members of the same and other ethnic groups. *Copyright © 1996 Elsevier Science Inc.* J CLIN EPIDEMIOL 49;12:1389–1399, 1996.

KEY WORDS. Pain, questionnaires, cross-cultural comparison, Mexican Americans, language, rheumatic diseases

INTRODUCTION

More than 17 million people aged 5 years and older in the U.S. speak Spanish at home, and 43.9% of these people speak English less than "very well" [1]. Thus, the Spanish speaking population of the U.S. is one of the largest in the world, larger than that of many Central and South American countries [2]. Not all U.S. Spanish speakers are immigrants; a sizable number were born in the U.S. and brought up speaking Spanish as their first language. Present demographic projections estimate that the number of Hispanics in the U.S. will increase further in the near future, with Hispanics becoming the second largest ethnic group by the mid-21st century [3]. In certain regions of the country, Hispanics are already the majority of the population [1]. Since many Hispanics speak Spanish, researchers wishing to measure health-related quality-of-life (HRQOL) in the U.S. will increasingly be faced with the challenge of adapting these measures, most of which were originally developed in English, to the Spanish language.

One such HRQOL scale is the McGill Pain Questionnaire (MPQ), developed by Melzack and Torgerson for the measurement of qualitative and quantitative aspects of pain [4,5]. Since it was published, the MPQ has been widely applied in the study of clinical [6–8] and experimental [9] pain. Because pain is a universal phenomenon in human experience, efforts to adapt the questionnaire to other languages quickly followed publication of the MPQ. To date we are aware of versions of the MPQ in Dutch [10], German [8,11], French [12,13], Italian [14,15], Finnish [16], Norwegian [17], and Arabic [18]. However, despite the worldwide distribution of the Spanish language and its increasing use in the United States, there are few published Spanish versions of the MPQ, and none of them provide empirical data related to their validity or reliability [19].

Our need for a Spanish version of the MPQ arose with the San Antonio Longitudinal Study of Aging (SALSA), a multidimensional, population-based, epidemiologic study of chronic disease and functional status in elderly Mexican-American and non-Hispanic white men and women. The MPQ was one of a number of HRQOL instruments translated into Spanish specifically for SALSA. In the present article, we describe the methods we followed for translating the MPQ into Spanish and present evidence of the equivalence between our Spanish translation and the original English version. Our methods closely parallel recently suggested guidelines for the cross-cultural adaptation of HRQOL instruments [20,21].

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One of the goals of cross-cultural research is to compare findings between different cultural groups. To accomplish this the measurement instruments used must be cross-culturally equivalent. Flaherty *et al.* [20] proposed a taxonomy of five dimensions that are important in establishing cross-cultural equivalence: (1) content equivalence, i.e., the extent to which concepts measured with the original language version are relevant to the culture of the target language; (2) semantic equivalence, i.e., the extent to which measurement items capture the same underlying meaning in the original and target language versions; (3) technical equivalence, i.e., the extent to which the mode of data collection has the same effect on the results obtained with different language versions in different cultural groups; (4) criterion equivalence, i.e., the extent to which different language versions of the measurement instrument are related to previously established and independent measures of the same phenomena; and (5) conceptual equivalence, i.e., the extent to which varying language versions of the instrument measure the same basic construct in the different cultural groups with which they are associated [20].

Our methods in the present study were designed to assure cross-cultural equivalence on the dimensions proposed by Flaherty *et al.* [20] and to conform to the guidelines more recently proposed by Guillemin *et al.* for the cross-cultural adaptation of HRQOL instruments [21]. Content equivalence was assured by having an expert panel of health care providers fluent in Spanish and knowledgeable about Mexican-American culture evaluate the relevance of each item of the MPQ to the culture of Spanish-speaking Mexican Americans. Semantic equivalence was ensured by using a rigorous forward- and back-translation process in which each MPQ item was first translated from the original English into the target language (Spanish) and then translated back to the original English. An integral part of the translation process was evaluation by an expert panel to assure comparability of item meanings across the two language versions. Technical equivalence was addressed in two ways. First, because Mexican Americans whose exclusive or primary language is Spanish often have little education and low literacy skills, we used an interviewer-administered method of data collection rather than paper-and-pencil self-administration for both language versions. Second, we compared the test–retest reliability of the two interviewer-administered language versions of the questionnaire. Criterion equivalence was addressed by comparing the two language versions of the MPQ against other established pain measures. Conceptual equivalence was addressed by comparing results obtained with the two language versions of the MPQ for patients with different clinical diagnoses and by comparing the association of the two language versions with other measured constructs theoretically linked to pain, e.g., health status measures.

METHODS

Translation and Evaluation Committees

Three separate groups of individuals participated in this cross-cultural adaptation process: one translation committee and two evaluation committees. The translation committee was formed by nine bilingual health researchers with extensive experience studying Mexican Americans. Spanish was the first language for five of the translators, three of whom were from Mexico, and one each from Panama and Spain. Among the four non-native Spanish speakers, two learned Spanish in Mexico and two in the U.S. All were residents of San Antonio and were familiar with local Spanish usage. Four of the translators were familiar with the concept and purposes of the MPQ. There were two evaluation committees: the first was composed of eight health-care providers: two native Mexican; three Mexican-American, and one Anglo-American physicians; one Anglo-American nurse; and one Mexican-American social gerontologist. Spanish was the first language for four of these individuals, but all were fluent and able to read and write in both languages, and had many years of clinical experience with Mexican-American patients. The second evaluation committee was formed by 10 bilingual health care consumers: five members of the housekeeping staff of our institution, one secretary, and one medical technician, all Mexican-American, one research assistant and one student, both Mexican; and one South-American housewife. Spanish was the first language for three of these people; all were fluent in both English and Spanish; and all were either born in or were long-term residents of South Texas.

Translation Technique

The members of the translation committee translated the English MPQ into Spanish working individually and in small groups, under the direction of one of us (HPH). A total of five iterations of the translation process were accomplished, including a formal back-translation. The original structure of the MPQ, which includes 78 single-word pain descriptors grouped into three categories (sensory, affective, and evaluative) and 20 subcategories (sensory: temporal, spatial, punctate pressure, incisive pressure, constrictive pressure, traction pressure, thermal, brightness, dullness, miscellaneous, tension, autonomic, fear, punishment, affective/evaluative, bother, radiation, tightness, coolness, and agony), was preserved. For each of the descriptors in English, a Spanish equivalent was found that was considered to convey similar qualitative and quantitative dimensions of pain.

Committee Review

The evaluation committee formed by health-care providers examined the appropriateness of the semantic content and grammatical form of each Spanish descriptor in the translation. This was done by providing committee members with lists of the translated pain descriptors, in which they were asked to rate the appropriateness of each word and, if indicated, to suggest alternatives. Differences were then resolved in a series of focus group meetings attended by committee members. The resulting Spanish MPQ is shown in the appendix.

Pain Intensity Content of the Spanish and English Pain Descriptors

One of the methods of reporting results of the MPQ is the Pain Rating Index (PRI), which is calculated by summing the ranks occupied by the single-word pain descriptors selected by subjects, within each pain subcategory. The rank of the pain descriptors within the subcategories was determined originally based on the pain intensity content of each word [4,5]. In the original English version, this was accomplished by asking volunteer psychology students, physicians, and patients to rate each word on a 7-point scale, based on the intensity of pain conveyed by each word. We emulated this process by asking the individual members of the two evaluation committees to rate the pain intensity content (PIC) of the English and Spanish single-word pain descriptor on a 100-millimeter visual analog scale (VAS). The members of each committee were provided with two lists of words, grouped by pain subcategory and with a 100-mm VAS

adjacent to the right of each descriptor, anchored by the words "No pain" on the left and "Worst pain" on the right. The order of administration of the Spanish and English word lists was randomized to preclude an ordering effect.

Testing of Spanish and English MPQs on Bilingual Patients

To compare results of the MPQ across languages controlling for inter-subject variation, we administered both versions of the questionnaire to 50 bilingual Mexican-American patients identified sequentially from a rheumatology clinic over a 9-week period. To be considered bilingual, the patients had to demonstrate the ability to speak both languages fluently and read out loud the items of a bilingual self-assessment questionnaire used in our clinic [22]. The diagnoses among the 50 patients included rheumatoid arthritis in 22; systemic lupus erythematosus in 11; fibromyalgia in 8; gout in 2; chronic low back pain in 2; osteoarthritis in 2; and 1 each of undifferentiated connective tissue disease, polymyositis, and shoulder impingement syndrome. The average age of patients was 44.8 years (range 20 to 73) and 10 were male. Average number of years of formal education was 10.2 (range 0 to 16). All of these patients were interviewed by a trained bilingual nurse who administered the questionnaire before and after the physician visit, alternating the order of the Spanish and English versions at random. The minimum amount of time between each version of the questionnaire was one hour. All patients had pain in the musculoskeletal system at the time of administration of the bilingual MPQ.

Reliability Testing in Monolingual Patients

To assess the test–retest reliability of the translated version of the MPQ compared to the English original, we studied 25 English-speaking and 25 Spanish-speaking, consecutively identified, monolingual in-patients who were hospitalized on general medical or surgical wards. These participants were selected for the study if they answered affirmatively when asked if they were having pain. Patients were interviewed in the language of their choice on two consecutive days by a trained bilingual nurse, during which the MPQ was administered. In addition, to compare reliability (i.e., stability over time) and concurrent validity (correlation with established measures of the same construct measured at the same time) of the two MPQ language versions, we included the following pain measurement instruments in the interviews: (1) Two 10-centimeter horizontal visual analog scales (VAS) for patients to rate the intensity of their pain at the time of the interview and during the past week; (2) the two MOS-SF36 "Bodily Pain" items [23]; (3) The McGill pain map [24]; and (4) the activities of daily living scale of the Modified Health Assessment Questionnaire (MHAQ) [25]. Spanish translations of each of these items are available and have been tested in South Texas Mexican Americans [23–25].

The causes of pain among the 50 hospitalized patients included traumatic injury without fracture in 13, elective surgery in 8, bone fracture in 5, superficial abscess in 4, gunshot wound in 3, amputation in 3, cellulitis in 2, pyelonephritis in 2, stroke in 2, osteoporosis in 2, and one each had AIDS, hepatic cirrhosis, kidney stone, systemic lupus erythematosus, aseptic meningitis, and cancer as the pain source. There were no significant differences between the Spanish- and English-speaking patients in their average age (43.1 [range 18 to 86] versus 37.5 [range 18 to 72], respectively, p = NS), or proportion of women (36% versus 39%, p = NS). However, Spanish-speakers were more likely to be Mexican-American (100% versus 60%, $p = 0.001$) and had significantly less formal education than the English-speakers (8.3 years [range 0 to 15] compared with 11.6 years [range 5 to 16], $p = 0.001$).

Analysis

Individual and average PIC ratings of the Spanish and English single-word pain descriptors were compared using Pearson correlation coefficients (r) [25]. We used a variety of approaches to measure agreement between the English and Spanish MPQs completed by the bilingual rheumatology patients and between the first and second questionnaires (test–retest reliability) completed by the monolingual in-patients. First, we calculated intra-class correlation coefficients (r_i) [26] between the English and Spanish questionnaires completed by bilingual patients or between the first and second questionnaires completed by monolingual patients. This resulted in an r_i value for each individual patient. We also measured agreement in the summary scores of the questionnaire, which in addition to the PRI, include the present pain index (PPI) and the number of words chosen (NWC). This resulted in an r_i value for each of these summary indices of the questionnaire.

In addition, since the MPQ provides a qualitative measure of pain among 20 subcategories included in the questionnaire, we measured agreement by subcategory.

This was done by considering each subcategory as a binary value depending on whether or not a word included in it was selected, and by considering the rank occupied by the selected word within the subcategory. Differences between these were tested using chi-square and paired t-tests, adjusting for multiple comparisons [27]. Agreement between subcategories was measured using Cohen's kappa [26] and r_i [28]. Since weighted kappa and r_i are considered to be equivalent [29], we interpreted both using published guidelines: kappa or r_i values ≥ 0.80 were considered to represent "almost perfect agreement"; values ≥ 0.75 were considered "excellent"; values ≥ 0.60 were interpreted as "substantial" agreement, and values ≥ 0.40 were considered "moderate" or "fair to good" [28,29]. Significance of observed differences in reliability was tested using Fisher z-transformed values for the correlation coefficients, as described [30]. Finally, the frequency of selection of each single-word pain-descriptor among bilingual patients was compared using chi-square [27].

RESULTS

Pain Intensity Content of Spanish Versus English Pain Descriptors

Figure 1 illustrates the two evaluation committees' VAS ratings of the PIC for each single-word pain descriptor. The individual rater and group mean scores for each single-word pain descriptor, given by the committee of health care providers, are shown in graphs A and C. Scores by the committee of health care consumers are shown in graphs B and D. Spanish and English PIC ratings correlated well with each other. Pearson correlations for scores given by each individual rater to each word were $r = 0.60$ for providers and $r = 0.58$ for consumers, Figure 1, A and B, respectively. The average PIC score given to each word by the raters as a group were $r = 0.85$ for providers and 0.80 for consumers, Figure 1, C and D. Correlations between average PIC scores for each word were strong between the two committees as well, with $r = 0.81$ for English descriptors and $r = 0.78$ for Spanish descriptors. There was no evidence for systematic

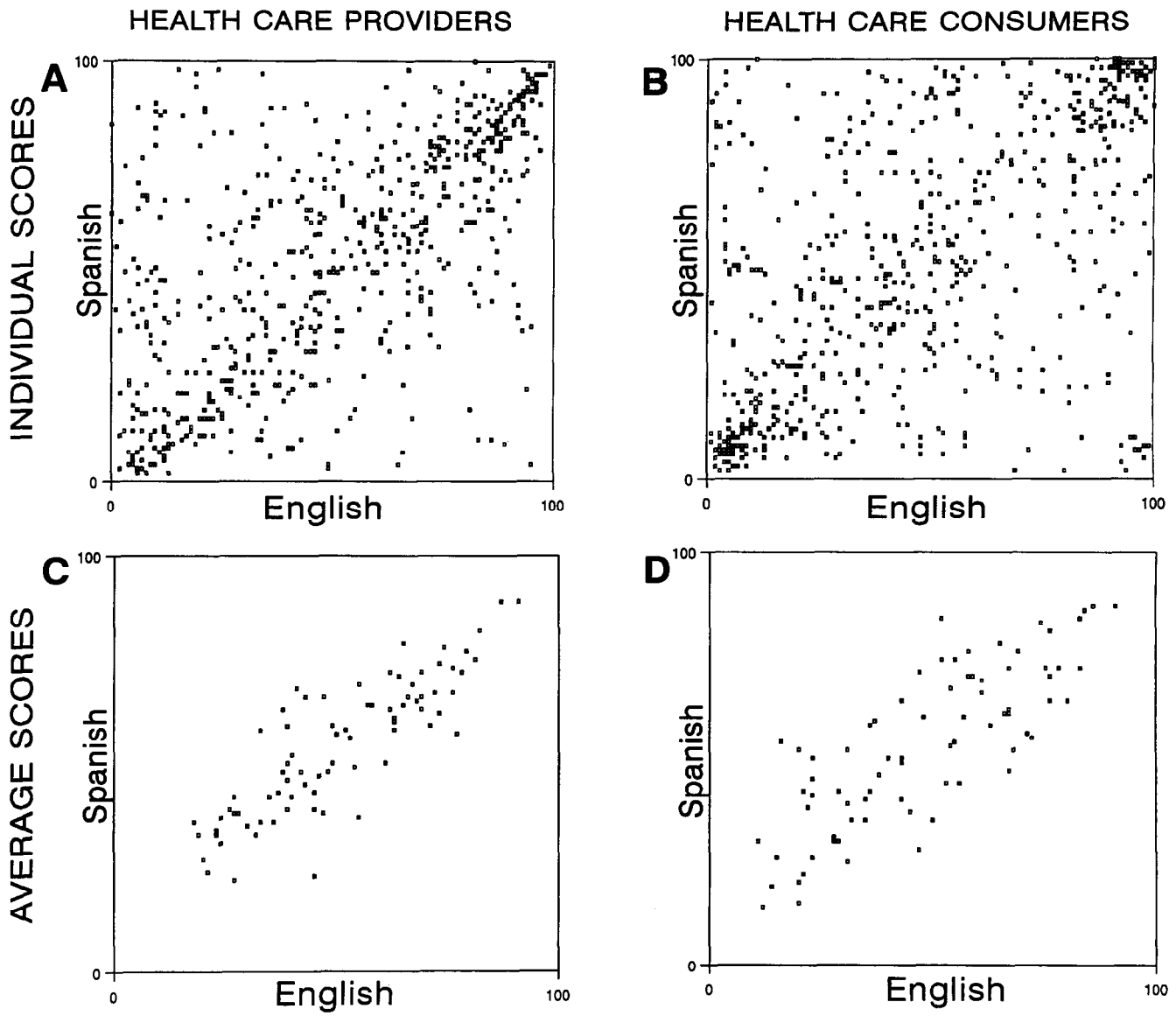


FIGURE 1. Scatter plots of the PIC ratings of Spanish and English pain descriptors. Members of a committee of bilingual health care providers (n = 8, graphs A and C) and a committee of health care consumers (n = 10, graphs B and D) were asked to rate the PIC of the pain descriptors in the MPQ in Spanish (Y axis) and English (X axis), on a 100-mm visual analog scale. In graphs A and B, each point represents the rating for one single-word pain descriptor by one committee member (r = 0.60 and 0.58). In graphs C and D, each point represents the average of the PIC scores given to a word by all the members of each committee (r = 0.85 and 0.80).

differences in PIC ratings between the two lists of pain descriptors. Thus, two independent groups of bilingual health care providers and consumers rated the PIC of the English and Spanish MPQ words as comparable.

Individual English–Spanish Comparisons in Bilingual Patients

Agreement between the ranks of the pain descriptors chosen by each patient in the Spanish and English MPQs are shown in Fig. 2. The r_i values were $\geq$ 0.60 in 24 of the patients (48%), signifying "substantial" or better agreement [28] while in 11 more patients $r_i \geq$ 0.40. Thus, agreement between the Spanish and English questionnaire in individual bilingual patients could be considered "fair to good" or higher in 35 of the patients, or 70% [26]. Using paired t-tests adjusted for multiple comparisons, there were no differences between Spanish and English MPQ word ranks in any of the patients. Patients with r_i higher than 0.60 were not different from those with r_i lower than 0.60 in terms of age, male:female ratio or educational level. However, the patient's diagnosis seemed to have an effect on the level of agreement between languages: the average r_i between Spanish and English among patients with fibromyalgia was lower than among patients with other conditions (Fig. 2).

Individual Test–Retest Reliability of MPQ Among Spanish- and English-Speaking Monolinguals

Reliability was $\geq$ 0.60, or "substantial," among 12 of the Spanish speakers (45%) and 13 of the English speakers (52%). Among three additional Spanish speakers and six English speakers, agreement be-

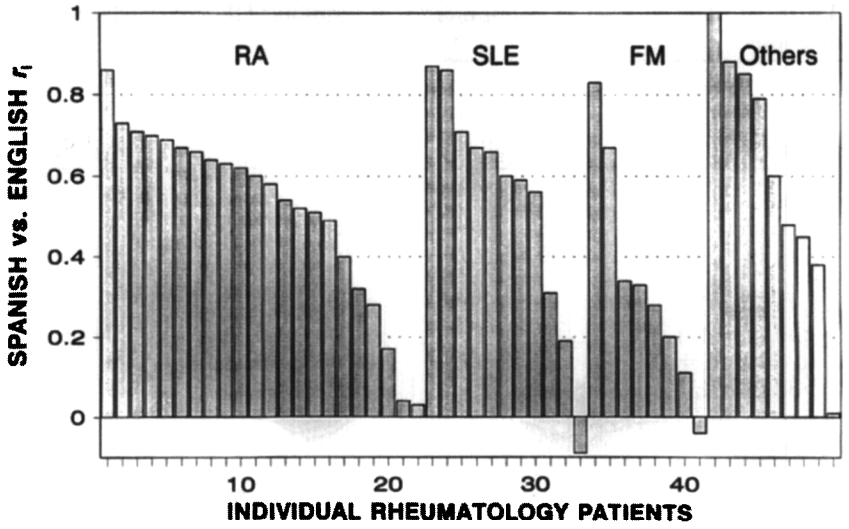


FIGURE 2. Agreement between the Spanish and English MPQ among 50 bilingual rheumatology patients who completed the questionnaire in both languages, grouped by diagnosis. *Bars* represent the value of r_i for each patient. Average r_i tended to be lower among the patients with fibromyalgia (average r_i = 0.34) than among patients with rheumatoid arthritis (RA, average r_i = 0.51), systemic lupus erythematosus (SLE, average r_i = 0.53) or other causes of musculoskeletal pain (average r_i = 0.60).

tween the first and second questionnaires was ≥ 0.40. Thus, reliability was "fair to good" or better in 60% of Spanish speakers and in 76% of English speakers (chi-square 0.86 with one degree of freedom, p = N.S.). As among the bilingual comparisons, we were unable to show an effect of age, gender, or years of education on the individual reliability of the MPQ, nor could we show an effect of the underlying diagnosis. Paired t-tests did not reveal significant differences between first and second MPQ among any of the Spanish- or English-speaking monolinguals.

English–Spanish Group Comparisons for Summary Indices of the MPQ in Bilingual Patients

Figure 3 illustrates the relationship between PRI (panel A) and NWC (panel B) for the English and Spanish MPQ. Mean PRI and NWC scores were not significantly different in Spanish and English (29.8 ± 14.7 versus 29.1 ± 15.8, p = 0.55 for PRI, and 12.9 ± 5.8 versus 12.4 ± 5.7, p = 0.07 for NWC), nor was there a significant difference in the PPI in either language (2.9 ± 1.4 versus 2.8 ± 1.2, p = 0.37). No significant difference between English and Spanish PRI and NWC was observed regardless of age, gender, educational level, or diagnosis (Table 1). Agreement between the two languages in the global MPQ scores was correspondingly "almost perfect" (r_i = 0.85 for PRI and 0.92 for NWC) (Fig. 3), and "fair to good" for the PPI (r_i = 0.55) [27,28]. In addition, the global agreement between Spanish and English word ranks of the full MPQ for the entire group was "substantial" (r_i = 0.60) [28].

Reliability and Concurrent Validity of the Summary Scores of the MPQ in Spanish- and English-Speaking Monolinguals

There were similar levels of agreement between the first and second administrations of the MPQ in both languages. In the case of the PRI, r_i = 0.72 in Spanish and 0.75 in English, a difference in correlation coefficients that is not statistically significant (z = 0.26, p >

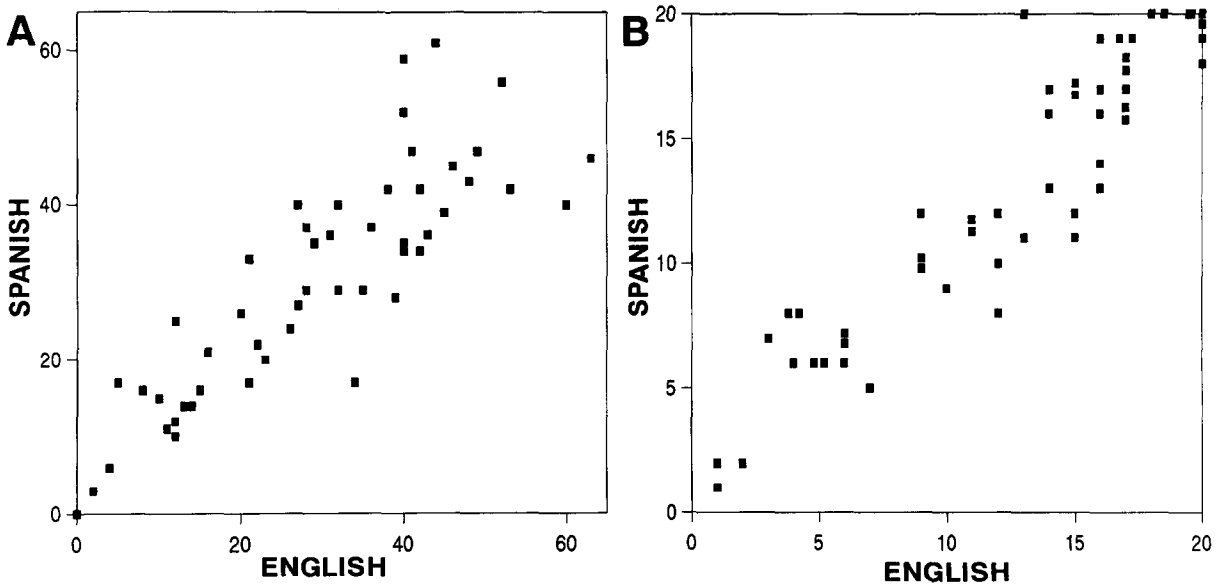


FIGURE 3. Scatter plots of Spanish and English pain rating index (A, r_i = 0.85) and number of words chosen (B, r_i = 0.92) among 50 bilingual rheumatology patients who completed the MPQ in both languages.

TABLE 1. McGill Pain Questionnaire scores among bilingual patients, stratified according to gender, age, education and diagnosis

	PRI (mean ± SD)		NWC (mean ± SD)		PPI (mean ± SD)	
Language	**English**	**Spanish**	**English**	**Spanish**	**English**	**Spanish**
Gender						
Women (n = 40)	28.8 ± 15.9	30.0 ± 15.2	12.3 ± 5.7	12.9 ± 5.5	2.8 ± 1.2	2.9 ± 1.3
Men (n = 10)	30.4 ± 16.0	28.6 ± 15.7	12.8 ± 6.2	12.9 ± 6.2	2.5 ± 1.4	2.8 ± 1.5
Age (years)						
<40 (n = 19)	30.1 ± 15.3	33.1 ± 15.8	13.2 ± 5.6	14.3 ± 5.6	2.8 ± 1.1	3.1 ± 1.4
40 to 49 (n = 18)	30.7 ± 17.4	29.5 ± 15.3	12.4 ± 5.8	12.4 ± 5.9	2.8 ± 1.1	3.0 ± 1.4
>50 (n = 13)	25.6 ± 14.8	25.4 ± 14.8	11.2 ± 6.0	11.7 ± 5.1	2.3 ± 1.0	2.6 ± 1.3
Education (years)						
0 to 8 (n = 13)	30.3 ± 13.7	31.3 ± 12.3	13.4 ± 5.0	14.6 ± 4.9	2.8 ± 1.3	3.1 ± 1.4
9 to 12 (n = 26)	31.3 ± 15.9	31.5 ± 14.4	13.2 ± 5.6	13.5 ± 5.6	2.7 ± 1.3	2.9 ± 1.4
≥13 (n = 11)	22.3 ± 17.2	23.7 ± 17.6	9.3 ± 5.8	9.6 ± 5.4	2.7 ± 1.3	2.7 ± 1.3
Diagnosis						
RA (n = 22)	28.0 ± 13.5	27.8 ± 13.0	12.1 ± 5.1	12.2 ± 5.1	2.5 ± 1.1	2.6 ± 1.1
SLE (n = 11)	26.1 ± 17.5	29.7 ± 17.3	11.3 ± 6.4	12.2 ± 6.1	2.7 ± 1.3	2.8 ± 1.6
FM (n = 8)	42.3 ± 16.3*	43.0 ± 10.3*	16.0 ± 5.3	17.3 ± 3.9	4.3 ± 1.0*	3.9 ± 1.0
Others[1] (n = 9)	24.2 ± 15.2	23.9 ± 14.3	11.6 ± 6.5	11.8 ± 6.5	2.1 ± 0.8	2.8 ± 1.7

Abbreviations: PRI = pain rating index, NWC = number of words chosen, PPI = present pain index, RA = rheumatoid arthritis, SLE = systemic lupus erythematosus, FM = fibromyalgia.

[1]See text for details.

*$p < 0.05$ compared to other diagnoses.

No significant differences between English and Spanish were observed.

0.05). Likewise, the reliability of the NWC was not significantly different between the two languages (r_i = 0.66 in Spanish and 0.74 in English, z = 0.84, p = NS for the NWC), nor was there a difference in the reliability of the PPI between languages (r_i = 0.62 in Spanish and 0.54 in English, z = 0.68, p = NS). Thus, the reliability of the summary scores of the translated MPQ was not significantly different from that of the English original. Summary scores of the MPQ Spanish and English versions, at both the first and second interviews, were correlated in the expected direction with the corresponding language versions of the other pain and health status measures, with 85% of the correlations in the moderate range (r = 0.26 to 0.55). Thus, the Spanish version of the MPQ also appears to have concurrent validity similar to the English original.

English–Spanish Comparisons for Pain Subcategories of MPQ in Bilingual Patients

As shown in Fig. 4, there were no significant differences in the proportions of bilingual patients endorsing words in any of the 20 subcategories in the Spanish and English versions. Similarly, there was no difference in the ranks of words selected in 19 of the 20 categories, with the only significant difference being in the subcategory of thermal pain (1.0 ± 1.1 vs 0.6 ± 0.6, p = 0.002, Fig. 4).

Agreement in the rank of the words selected was "fair to good" or better in 15 of the 20 subcategories of the questionnaire, and "fair" in the remaining 5 [26,28]. When each subcategory was considered categorically as positive or negative if one of their words was selected or not, agreement was "fair to good" or better in all ($\kappa \geq$ 0.40) and "substantial" in 10 of the 20 ($\kappa \geq$ 0.60) [26]. There was near perfect agreement in the average score for each subcategory between English and Spanish (r_i = 0.92).

Reliability of the MPQ in Spanish- and English-Speaking Monolinguals According to Pain Subcategories

The test–retest reliability of the individual items of the MPQ, represented by the subcategories of the questionnaire, was ≥ 0.40 in 10 (50%) in Spanish and in 11 (55%) in English. Reliability in the "substantial" range was reached only in 4 of the 20 subcategories, in either language. Comparisons of the ranks of words selected according to each sub-category using paired t-tests revealed no significant differences between the first and second interviews, in either language.

Reliability of the MPQ Compared with Other Measures of Pain or Health Status

Table 2 shows the test–retest reliability of the different pain measurement instruments administered to the Spanish- and English-speaking monolingual patients. The intraclass correlation coefficients for the indices derived from the MPQ in Spanish were not significantly different from those obtained with the original English questionnaire. Significant differences in reliability between languages were observed only for the MOS-SF36 pain-interferes with work item, in which reliability was significantly higher in Spanish (Table 2). Notably, the intraclass correlations for the MPQ scales compared well in both languages to the other scales included in the interviews.

Frequency of Selection of Individual Words by Bilingual Patients

Figure 5 shows the frequency of selection by bilingual patients of each of the English and Spanish words in the MPQ. All of the En-

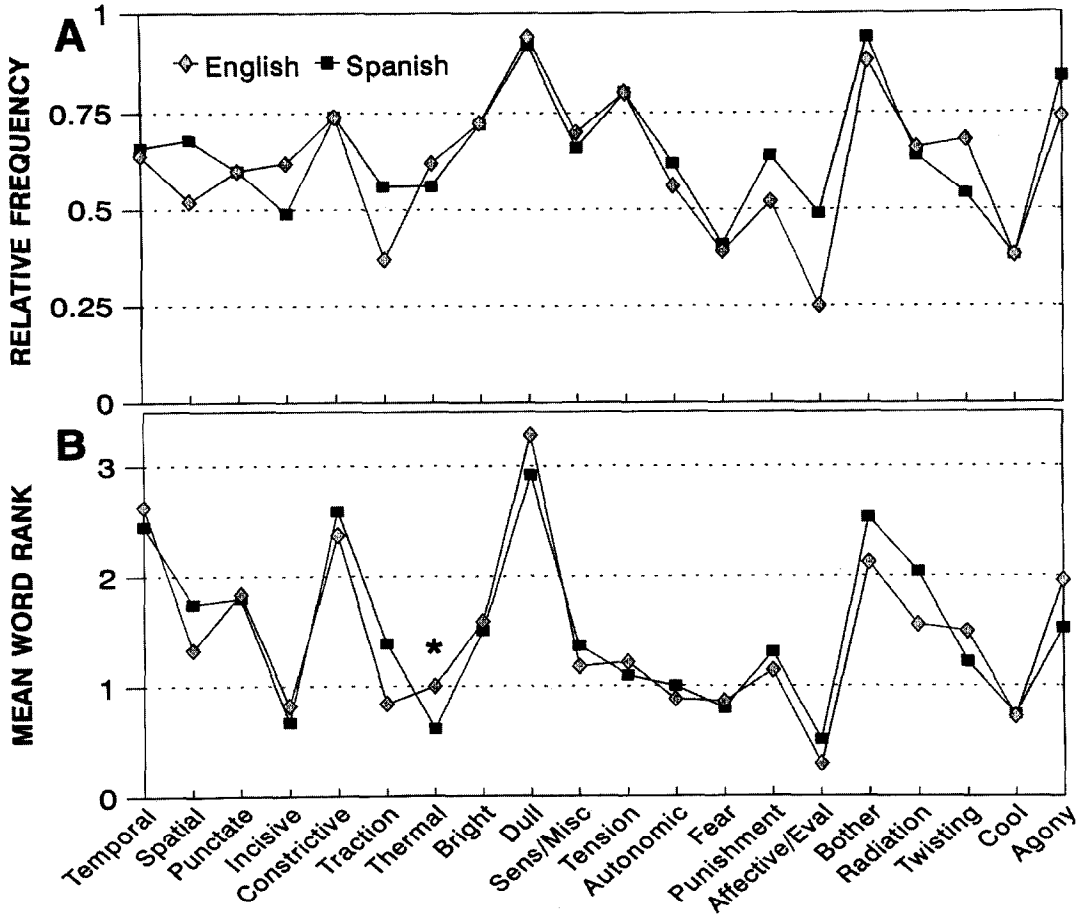


FIGURE 4. (A) Proportion of the 50 bilingual patients selecting words within pain subcategories. There was no difference between Spanish and English in any of the subcategories ($\chi^2 < 3.78$ with 1 degree of freedom for all pair-wise comparisons). (B) Mean rank of the words selected within each subcategory. After adjusting for multiple comparisons, only the average rank of words describing thermal qualities of pain was different between Spanish and English (*$p = 0.0023$, two-tailed, paired *t*-test).

TABLE 2. Test–retest reliability of pain measurement instrument in English- or Spanish-speaking monolingual patients

Pain measure	Reliability in English[a]	Reliability in Spanish[a]	*p*-value
Pain rating index[b]	0.75	0.72	NS
Number of words chosen[b]	0.66	0.74	NS
Present pain index[b]	0.54	0.62	NS
Number of painful areas[c]	0.77	0.73	NS
Pain now VAS	0.69	0.38	NS
Pain past week VAS	0.70	0.51	NS
Bodily pain[d]	0.74	0.83	NS
Pain interferes with work[d]	0.50	0.83	<0.05
MHAQ[e]	0.62	0.70	NS

Abbreviations: NS = not significant, VAS = visual analog scale.
[a]Reliability was measured by the intraclass correlation coefficient between the first and second interviews, conducted within one day of each other.
[b]From the McGill Pain Questionnaire [4].
[c]From the McGill Pain Map [24].
[d]From the MOS-SF36 [23].
[e]Modified Health Assessment Questionnaire [25].

glish words in the questionnaire were selected by at least one of the patients with musculoskeletal pain. The English words most often chosen by the patients to describe their pain were *tender*, *sharp*, *aching*, *exhausting*, *tiring* and *annoying*, each selected by 40% or more of the patients. These words belong in the "sensory-miscellaneous," "incisive pressure," "dullness," "tension," and "evaluative" subcategories of the MPQ. By comparison, three of the Spanish words were not selected by any of the 50 bilingual patients, although it should be noted that their corresponding English counterparts were selected by only two patients each (p = N.S.). The most often chosen Spanish words were *fastidioso* (nagging), *fatigante* (tiring), *caliente* (hot), *molesto* (troublesome), *punzante* (shooting), and *afligido* (wretched). In Spanish, these words fall into the "tension," "thermal," "evaluative," "spatial," and "autonomic" subcategories of the MPQ.

There were significant differences in the rate of endorsement of 10 of the 78 words in the questionnaire. These included the English–Spanish word pairs *taut–tenso* (4% vs 26%), *piercing–punzante* (10% vs 24%), *troublesome–molesto* (10% vs 44%), *wrenching–torciente* (16% vs 38%), *spreading–esparciente* (10% vs 4%), *wretched–afligido* (22% vs 48%), *throbbing–palpitante* (30% vs 12%), *nagging–fastidioso* (32% vs 60%), *annoying–enfadoso* (40% vs 16%) and *tender–delicado*

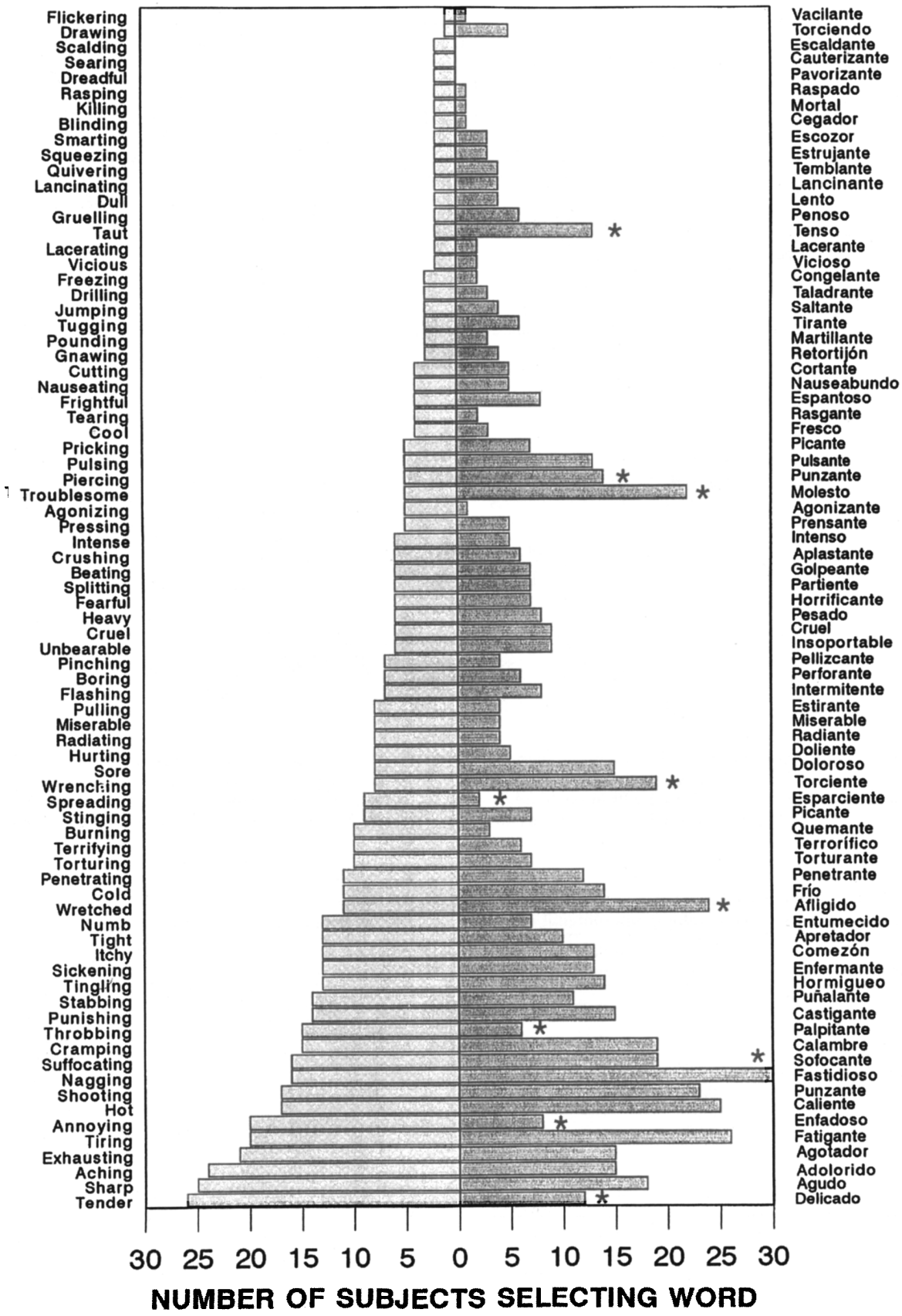


FIGURE 5. Frequency of endorsement of each of the individual words in the McGill Pain Questionnaire. *Horizontal bars* correspond to the number of bilingual patients who selected each word. Values to the left of 0 correspond to the English version and values to the right correspond to the Spanish version of the words. Significant differences in the rate of endorsement were observed for 10 of the 78 words, 4 of which were endorsed more frequently in English and 6 more frequently in Spanish. (*$\chi^2 \geq 3.84$ with 1 degree of freedom, $p \leq 0.05$ for each pair-wise comparison. Differences in 4 of the 78 comparisons are expected due to chance alone).

(52% vs 24%). There was no difference in the rate of endorsement in the remaining 68 single-word pain descriptors (Fig. 5).

DISCUSSION

We have adapted the MPQ to the Spanish language spoken by Mexican Americans in South Texas and presented evidence of its cross-cultural equivalence to the original English version. Although there was some disagreement between languages with individual patients, in aggregate these discrepancies were randomly distributed. Thus, comparing groups of persons with pain, there was an excellent agreement between the Spanish and English versions of the MPQ [29] and similar levels of reproducibility. Our methods address the various validation dimensions proposed by Flaherty *et al.* [20] and conform to the guidelines that were recently suggested by Guillemin *et al.* [21] for the cross-cultural adaptation of the HRQOL measures:

(1) We employed a panel of translators from a variety of backgrounds, most of them translating into their mother tongue (Spanish for the forward-translation, English for the back-translation), some of whom were aware of the concept and objectives of the questionnaire. The Spanish version of the MPQ was refined to reflect the pattern of Spanish utilized in South Texas through multiple iterations of forward- and back-translations, with a series of evaluations of the relevance of items to the target culture and of the comparability of meaning of the Spanish and English items.

(2) We assembled an expert committee to evaluate the Spanish translation, reach consensus on individual words, and verify the *content* and *semantic* equivalence of the final Spanish versions with the original English MPQ [20,21].

(3) Using independent committees of health-care providers and consumers, we evaluated the PIC of each single-word pain descriptor in both language versions of the MPQ. This step is analogous to the *weighting scores* guideline of Guillemin *et al.* [21];

(4) We further demonstrated semantic equivalence of the Spanish and English MPQ by administering them to bilingual patients suffering from musculoskeletal pain; as suggested by Guillemin *et al.* [21]

(5) We showed that the reliability of the Spanish MPQ is equivalent to the reliability of the original English among monolingual patients when both language versions are interviewer-administered, and that their relationship with concurrently obtained measures is similar. This step provides evidence for the *technical* equivalence of the two language versions [20].

(6) We demonstrated that the reliability of both the Spanish and English MPQ is comparable to the reliability of other recognized measures of pain, which provides evidence of their *criterion* equivalence.

(7) We demonstrated differences in questionnaire results between patients with different types of pain in both languages. We also showed that the two language versions were similarly correlated with measures of functional status. These steps provide evidence for the Spanish and English MPQ's *conceptual* equivalence [20].

By testing the questionnaire among clinical samples of patients with pain due to rheumatic, post-operative, traumatic, and other acute medical conditions, we have ensured that different types of pain were represented. This can be observed graphically in Figure 3, where the distribution of pain intensity ranges from a PRI of 0, the lowest possible score, to a high of 62 out of a possible 78. All of the subcategories of pain were endorsed, which is shown graphically for the bilingual patients in Figure 4, and all but three of the Spanish words were selected by at least one of the patients in this study (Fig. 5).

Agreement between the Spanish version we have adapted and the original English MPQ can be considered "near perfect" or "excellent" for the global scores of the questionnaire [26,29]. This means that we can now undertake studies comparing pain intensity between Spanish- and English-speaking subjects with a high level of confidence that the results obtained will be independent of language (31). Agreement was more variable within individual patients, ranging from no agreement at all to perfect agreement (Fig. 2). Thus, while suited for the study of groups, caution must be used when applying the MPQ as a clinical tool for comparing individual subjects. Comparisons of the quality of pain between Spanish and English, although less efficient than comparisons of pain intensity, are also possible with our Spanish version of the questionnaire, as evidenced by the similar frequency of selection of words within each subcategory and the similar rank of words selected (Fig. 4). Moreover, agreement was "fair to good" in the majority of the subcategories.

The reproducibility of the MPQ summary indices, the PRI and the NWC, is good in both languages (Table 2). However, as is often the case with multi-item questionnaires, the reliability of the individual items is lower than that of the whole instrument. In the MPQ, each of the pain subcategories as well as the PPI is equivalent to an individual item. Thus their notably lower English–Spanish agreement and test–retest reliability, compared to the summary scores, is not surprising. Thus, caution is advised for the interpretation of results along the individual subcategories of the questionnaire. We wish to emphasize that the shortcomings of the MPQ subcategories in terms of reproducibility apply to the original English as well as to the translated Spanish version.

We did not examine the responsiveness of this instrument to clinical change. However, a pre-requisite for responsiveness to change is stability. We considered that an interval of 24 hours between the first and second interviews in the reliability study would offer the best balance to minimize recall of previous answers and the occurrence of clinically significant change in the amount and type of pain. Our findings show that both language versions of the questionnaire provide stable results among patients with various acute and chronic causes of pain. Further study is needed to examine the responsiveness of the Spanish version of the questionnaire.

In spite of the small number of bilingual patients included, both versions of our questionnaire detected significantly higher levels of pain, measured by the PRI, among patients with fibromyalgia compared with those with other types of rheumatic pain. Trends toward higher values in the NWC and PPI were also observed among fibromyalgia patients, corresponding to the findings of others [32]. Although measuring the intensity of pain in different conditions was not a main objective of the study, this finding strengthens the validity of both versions of the questionnaire.

One Spanish translation of the MPQ has been previously published. However, neither the translation process utilized nor the equivalence and validity of that version were reported [19]. A reconstructed, Argentine–Spanish version of the questionnaire is described in abstract form [33], but we were unable to find a publication containing the actual questionnaire. A Spanish version translated and validated in Mexico exists, but remains unpublished [34]. Finally, we were recently forwarded a Spanish questionnaire that is being employed to study non-articular rheumatism in Spain (Dr. Mulero Mendoza, Hospital Puerta de Hierro, Madrid, Spain, *personal communication*).

It is instructive to compare the four Spanish versions of the MPQ available to us [19,34]. Table 3 shows the words included in the

TABLE 3. Four Spanish language versions of the McGill Pain Questionnaire compared (subcategory of constrictive pressure)

English [4]	South Texas[a]	Spain [19]	Mexico [33]	Spain[b]
Pinching	Pellizcante	Pellizco	Pellizcador	Entumecimiento
Pressing	Prensante	Opresión	Apretador	Como un pellizco
Gnawing	Retortijón	Roedor	Prensador	Agarrotamiento
Cramping	Calambre	Calambre	Estrujador	Calambre
Crushing	Aplastante	Estrujamiento	Retorcedor	Espasmo
—	—	—	Triturador	Retortijón Opresivo

[a]Present study.
[b]M Mendoza (personal communication).

subcategory of constrictive pressure type pain. There was only uniform agreement between the four versions in one of the words (pinching); but the grammatical form of the word differs in all four. More importantly, there is considerable disagreement between the Spanish versions in all the remaining words, and two of them have more than the 5 words in the original English version. Similar differences between the Spanish versions were present in the remaining 19 pain subcategories. Thus, it is likely that considerable variation in the results would arise between studies using each of these three Spanish versions in the same population. This illustrates the pitfalls of studying Spanish-speaking populations with instruments translated from English without first evaluating their performance in the target population. Researchers should keep in mind that, as shown in our brief example in Table 3, the Spanish vernacular and resulting translations can vary considerably from one geographic location to the next. In the U.S., distinct populations of Spanish speakers exist in New York, who mostly originate from Puerto Rico and the Dominican Republic; South Florida, mostly from Cuba; and South Texas, New Mexico, Arizona and California, mostly from Mexico. It is possible, if not likely, that substantial differences in the Spanish spoken may exist within these regions and that a single Spanish version of the MPQ will not be suitable for all Hispanic subgroups. Thus, prior to applying any translated HRQOL measure to Spanish-speaking subjects in the U.S., particularly Hispanics not of Mexican origin, preliminary fieldwork to test its performance, validity, and cross-cultural equivalence to the original English in the intended target population should be completed. We urge researchers who work with other Hispanic subgroups to test the generalizability of the Mexican American Spanish MPQ to their target populations and to make any additional cross-cultural adaptations that may be required.

In summary, we have cross-culturally adapted the MPQ into the Spanish language used in South Texas and provided evidence that the translation fulfills published criteria for cross-cultural equivalence. The Spanish MPQ presented here is suitable for studying Spanish-speaking Mexican Americans in South Texas, and probably in other locations in the Southwestern U.S. However, as outlined in the preceding paragraphs, preliminary fieldwork to assess the performance of the questionnaire is necessary prior to widespread application in other populations.

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APPENDIX 1. Spanish McGill Pain Questionnaire (San Antonio version)

Pain subcategory				
Temporal	**Spatial**	**Punctate pressure**	**Incisive pressure**	**Constrictive pressure**
Vacilante	Saltante	Picante	Agudo	Pellizcante
Temblante	Intermitente	Perforante	Cortante	Prensante
Pulsante	Punzante	Taladrante	Lacerante	Retortijón
Palpitante		Puñalante		Calambre
Golpeante		Lañcinante		Aplastante
Martillante				
Traction pressure	**Thermal**	**Brightness**	**Dullness**	**Sensory: misc.**
Tirante	Caliente	Hormigueo	Lento	Delicado
Estirante	Quemante	Comezón	Doloroso	Tenso
Torciente	Escaldante	Escozor	Doliente	Raspado
	Cauterizante	Picante	Adolorido	Partiente
			Pesado	
Tension	**Autonomic**	**Fear**	**Punishment**	**Affective: evaluative**
Fatigante	Enfermante	Horrificante	Castigante	Afligido
Agotador	Sofocante	Espantoso	Penoso	Cegador
		Terrorifico	Cruel	
			Vicioso	
			Mortal	
Bothersome	**Radiation**	**Tightness**	**Coolness**	**Agony**
Enfadoso	Esparciente	Apretador	Fresco	Fastidioso
Molesto	Radiante	Entumecido	Frío	Nauseabundo
Miserable	Penetrante	Torciendo	Congelante	Agonizante
Intenso	Punzante	Estrujante		Pavorizante
Insoportable		Rasgante		Torturante